

Code Explanation

Code Explanation: `glue\_pyspark/src/main/\_\_init\_\_.py`

Purpose

This is a **Python package initialization file** for an AWS Glue PySpark ETL job. It exposes the main functions from the `job` module to external callers.---

Step-by-Step Breakdown

1. **Module Docstring**

```
"""Main package for AWS Glue PySpark job implementationContains core ETL logic and transformations"""
```

- Documents the package's purpose: housing ETL logic for AWS Glue jobs using PySpark.---

2. **Import Statement**

```
from .job import ( initialize_spark_contexts, read_data_safe, write_data_safe,
validate_s3_path, clean_data, apply_scd_type2, calculate_customer_aggregate_spend,
main)
```

- **Imports 8 functions** from the sibling `job.py` module (relative import via `job`).- These functions represent the core ETL operations: - **`initialize\_spark\_contexts`**: Sets up Spark/Glue contexts. - **`read\_data\_safe`**: Reads data with error handling. - **`write\_data\_safe`**: Writes data with error handling. - **`validate\_s3\_path`**: Validates S3 URI format/accessibility. - **`clean\_data`**: Performs data cleansing transformations. - **`apply\_scd\_type2`**: Implements Slowly Changing Dimension Type 2 logic. - **`calculate\_customer\_aggregate\_spend`**: Computes customer-level aggregations. - **`main`**: Entry point for the Glue job.---

3. **Public API Definition**

```
__all__ = [ "initialize_spark_contexts", "read_data_safe", "write_data_safe",
"validate_s3_path", "clean_data", "apply_scd_type2",
"calculate_customer_aggregate_spend", "main"]
```

- **`\_\_all\_\_`** explicitly declares which symbols are exported when someone does:

```
from glue_pyspark.src.main import *
```

- Restricts the public interface to these 8 functions, hiding internal implementation details.---

Summary

This file acts as a **controlled gateway** to the ETL job's functionality:- Centralizes imports from `job.py`.- Defines a clean public API surface.- Enables external scripts to call these functions without directly importing from `job.py`.